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Appendix

A Baseline result of GSM8K

We compare the accuracy result of GSM8k for the subset of 300 samples and the full set of 1319 samples in the four baselines we used. The result is shown below, which ensures that the randomly selected data samples can roughly represent the entire dataset.

	Subset	Full Set
Llama3.1-8B-it	60.33	54.74
DeepSeek-Llama3-8B	82.67	82.26
DeepSeek-distill-Qwen-1.5B	70.00	70.20
Gemma-2-9b-it	88.00	87.79

Table 3: Accuracy of subset and full set of GSM8K in four LLMs

B Feature Selection Details

In the experiment of feature selection, we found that for `llama_scope_r1_distill(l15r_800m_openr1_math)`, the features obtained by $\alpha_t^q = \frac{1}{N}\sum_{k=1}^{N}\|\alpha_t^{x_k} - \alpha_t^{y_k}\|$ contains many noise features, and we are very difficult to obtain useful features from this equation. This may be related to the training corpus. This SAE was only trained on one corpus. We did not find that this phenomenon exists in other SAEs with richer training corpora such as `gemma-scope-9b-it-res(layer_9/width_16k/canonical)` and `llama_scope_lxr_8x(l15r_8x)`. It confirms our assumption. So we leverage $\alpha_t^q = \frac{1}{N}\|\sum_{k=1}^{N}(\alpha_t^{x_k} - \alpha_t^{y_k})\|$ to replace the original equation to obtain the final features in `llama_scope_r1_distill(l15r_800m_openr1_math)`.

C Steering Parameter Recommendations

For SAE steering, we found features with the following indexes: 24715,20737,20236,14276,15593, and 17831 are suitable features for steering in `llama_scope_r1_distill(l15r_800m_openr1_math)`. Features with index 13419,12085, and 9540 are suitable for `gemma-scope-9b-it-res(layer_9/width_16k/canonical)`. After experiments, it was found that large strength would cause the LLM's generation to be disordered. Therefore, we recommend strength $\lambda \leq 0.5$.

D SAE-Free Experiment Implementation Details

For SAE-free steering, we only conduct experiments on the eigenvectors corresponding to the top-10 eigenvalues, and the strength λ is still recommended to be less than 0.5.

For the Booststep baseline, most of our experimental settings are the same as the original settings. To speed up the generation efficiency, we set the maximum number of sub-steps to 10, the similarity retrieval threshold to 0.7, the maximum number of tokens in each sub-step to 200, and the loop will be exited when the answer contains the word "final answer" or the maximum number of steps is generated. For the Mathneuro baseline, we randomly sample 1000 samples of `gsm8k.csv` and `race.csv` provided by the original paper to calculate the active neurons for math problems and non-math problems, and set the keep ratio to 15% and the activation multiple to 1.1.

E Derivation of Formula 7

Based on the Sparsity Assumption, we have $\hat{d} < d_{\text{model}}$, which means it's feasible to find $\hat{d}$ mutually orthogonal unit vectors in d_{model}-dimensional space. Then according to Orthogonality Assumption, we have $\langle \boldsymbol{z}_i, \boldsymbol{z}_j \rangle = 0$ when $i \neq j$, so:

$$\|\langle \boldsymbol{x}_p - \boldsymbol{y}_p, \boldsymbol{z}_t \rangle\| = \|\langle \sum_{i=1}^{\hat{d}} \boldsymbol{z}_i \left(\alpha_i^{\boldsymbol{x}_p} - \alpha_i^{\boldsymbol{y}_p}\right), \boldsymbol{z}_t \rangle\|$$
$$= \|\sum_{i=1}^{\hat{d}} \left(\alpha_i^{\boldsymbol{x}_p} - \alpha_i^{\boldsymbol{y}_p}\right) \langle \boldsymbol{z}_i, \boldsymbol{z}_t \rangle\| = \|\left(\alpha_t^{\boldsymbol{x}_p} - \alpha_t^{\boldsymbol{y}_p}\right)\| \tag{9}$$

For problem 7, it is a Rayleigh quotient problem. We use the Lagrange multiplier method to solve the above problem.

$$L(u, \lambda) = z^\top \boldsymbol{A}\boldsymbol{A}^\top z - \lambda(z^\top z - 1) \tag{10}$$

Therefore

$$\frac{\partial L}{\partial z} = 2\boldsymbol{A}\boldsymbol{A}^\top z - 2\lambda z = 0, \frac{\partial L}{\partial \lambda} = z^\top z - 1 = 0$$

So we have

$$2\boldsymbol{A}\boldsymbol{A}^\top z - 2\lambda z = 0$$
$$\Longrightarrow \boldsymbol{A}\boldsymbol{A}^\top z = \lambda z \Longrightarrow \max_{\|z\|=1} z^\top \boldsymbol{A}\boldsymbol{A}^\top z$$
$$= \max_{\|z\|=1} \lambda z^\top z = \lambda_{max}$$